

MODBUS RTU Servo Communication Parameters:

Station number setting : PA_000, **Baud rate setting** : PA_00D, Default is 19200.

Data Bits : 8 bits, **Parity**: Even, **Stop Bits**: 1 (These communication parameters are fixed and cannot be modified).

MODBUS RTU Communication Address Conversion Instructions:

Example: PA_190 (Internal Position 0 Speed) corresponds to the hexadecimal value 190, which is a decimal address of 400 (When communicating with some PLCs or touch screens, the address must be incremented by 1).

Position Mode Control Description

1. Servo basic parameter settings (after the configuration is completed, a restart is required to take effect. It is recommended to modify these parameters using the debugging software)

Parameter number	Parameter name	Setting value	Explanation
PA_002	Control mode selection	3	Select position/speed switching mode
PA_004	Forward and reverse limit mode selection	0	For limit mode, refer to PA_066 parameter settings.
PA_058	Acceleration time setting	200	Set the startup acceleration time
PA_059	Deceleration time setting	200	Set the stop deceleration time
PA_090	Extended mode selection	1	Select extended mode (communication control)
PA_094	Jet position control selection	0	Select absolute position control mode
PA_096	Position device mode selection	2	Rising edge loading
PA_084	DI4 function configuration	4	Control mode switching
PA_085	DI5 function configuration	20	Position device signal
PA_086	DI6 function configuration	17	Origin switch
PA_087	DI7 function configuration	16	Return to zero start
PA_08F	Servo enable selection	1	Servo automatically enabled when powered on

Modify and save the above parameters through the servo panel or debugging software, and then restart the servo.

2. Communication control parameter description

2.1 Common communication parameters

Parameter number	Correspondence address (decimal)	Parameter name	Explanation
PA_1A4	420	Analog IO control	Control the status of each DI port using parameter values
PA_1D3	467	DO status monitoring	Monitor the current servo status
PA_1B8	440	Command position	Command position set by the host computer
PA_1B9	441		
PA_1BC	444	Feedback position	Actual position feedback from the motor
PA_1BD	445		

2.2 Internal position parameter table

Parameter number	Correspondence address (decimal)	Parameter name	default value	Function and meaning
PA_168	360	Internal position command 0	0	0th internal position command
PA_169	361			
PA_190	400	Internal position 0 speed	0	
PA_16A	362	Internal position command 1	0	1st internal position command
PA_16B	363			
PA_191	401	Internal position 1 speed	0	
PA_16C	364	Internal position command 2	0	2nd internal position command
PA_16D	365			
PA_192	402	Internal position 2 speed	0	
PA_16E	366	Internal position command 3	0	3rd internal position command
PA_16F	367			
PA_193	403	Internal position 3 speed	0	
PA_170	368	Internal position command 4	0	4th internal position command
PA_171	369			
PA_194	404	Internal position 4 speed	0	
PA_172	370	Internal position command 5	0	5th internal position command
PA_173	371			
PA_195	405	Internal position 5 speed	0	
PA_174	372	Internal position command 6	0	6th internal position command
PA_175	373			
PA_196	406	Internal position 6 speed	0	
PA_176	374	Internal position command 7	0	7th internal position command
PA_177	375			
PA_197	407	Internal position 7 speed	0	
PA_178	376	Internal position command 8	0	8th internal position command
PA_179	377			
PA_198	408	Internal position 8 speed	0	
PA_17A	378	Internal position command 9	0	9th internal position command
PA_17B	379			
PA_199	409	Internal position 9 speed	0	
PA_17C	380	Internal position command 10	0	10th internal position command
PA_17D	381			
PA_19A	410	Internal position 10 speed	0	
PA_17E	382	Internal position command 11	0	11th internal position command
PA_17F	383			
PA_19B	411	Internal position 11 speed	0	
PA_180	384	Internal position command 12	0	12th internal position command
PA_181	385			
PA_19C	412	Internal position 12 speed	0	
PA_182	386	Internal position command 13	0	13th internal position command
PA_183	387			
PA_19D	413	Internal position 13 speed	0	

PA_184	388	Internal position command 14	0	14th internal position instruction
PA_185	389			
PA_19E	414	Internal position 14 speed	0	
PA_186	390	Internal position command 15	0	15th internal position instruction
PA_187	391			
PA_19F	415	Internal position 15 speed	0	

3. Communication control DI port description

3.1、PA_1A0 and PA_1A4 control DI port description

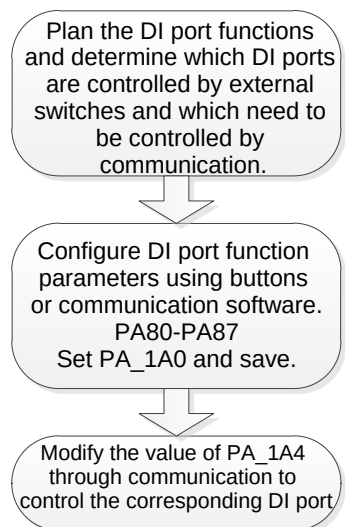
Parameter number	Parameter function	Parameter value binary bit status							
		DI7	DI6	DI5	DI4	DI3	DI2	DI1	DI0
PA_1A0	Word address corresponding bit	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	External IO/analog IO switching	0	1	0	1	0	0	1	0
	When the corresponding bit is set to 0, the corresponding DI port is controlled by external wiring. When set to 1, the corresponding DI port is controlled by analog IO, and the control parameter is PA_1A4.	External Control	Communication Control	External Control	Communication Control	External Control	External Control	Communication Control	External Control
PA_1A4	Word address corresponding bit	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	Communication analog IO	0	0	0	1	0	1	1	0
	When the corresponding bit of PA_1A0 is set to 1, this parameter can modify the status of the corresponding DI port. Setting it to 1 means that the DI port is valid.	External Control	DI disconnected	External Control	DI connected	External Control	External Control	DI connected	External Control

Example: In the table above, DI1, DI4, and DI6 need to be set to communication control, and other DI ports are conventional external IO control. Then the binary value of PA_1A0 is 01010010, which is converted to decimal as 82. If all DI ports are to be set to communication control, PA_1A0 is set to 255.

The DI port control steps are shown in the figure on the right:

After PA_1A0 is set, you can control the DI port's on/off state by modifying the PA_1A4 parameter value. For example, in the table above, to turn DI1 on, set Bit 1 of PA_1A4 to 1. Assuming the drive station number is 1, the communication control steps are as follows:

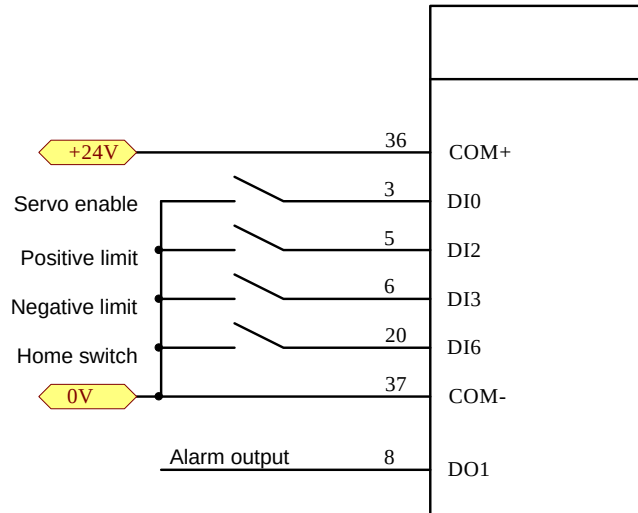
- 1) Read the PA_1A4 value using the following communication message: 01 03 01 A4 00 01 C4 15;
- 2) Assuming the read value is 128, convert it to binary 10000000;
- 3) Perform a bitwise OR operation on the PA_1A4 value (10000000) and the binary value (10), resulting in a value of 10000010;
- 4) Convert 10000010 to decimal to 130;
- 5) Write 130 into PA_1A4 using the following communication message: 01 06 01 A4 00 82 49 B4;



4. Communication Control Servo Function Description

4.1. Return to zero function description

- ① First, connect the forward/reverse limit switch and origin switch as shown below
(if PA_8F is set to 1, DI0 does not need to be connected)



- ② After the servo is powered on, the PA_1D3 parameter value can be detected (DO status monitoring parameter. For example, if DO0 (Bit 0 of PA_1D3) is 1, it means that the servo is ready and can start controlling the return to zero action).
- ③. Start zero return (set Bit7 of PA_1A4 to 1). At this time, the motor will perform zero return action, detect the origin switch, reverse, find the Z signal and then stop. At this time, the zero return action is completed and the current position of the servo will be cleared.

4.2 Forward and reverse jog function description

- ① Switch the mode to speed mode via DI4 (set Bit4 of PA_1A4 to 1), and set the speed values of internal speed 1 and internal speed 2 as follows:

Parameter number	Description	Setting value
PA_141	Internal speed command 1	100
PA_142	Internal speed command 2	-100

- ② Select the speed using the speed index parameter PA_092:
When set to 1, the motor rotates forward at a speed of 100;
When set to 2, the motor rotates reverse at a speed of -100;
When set to 0, the motor stops.
To switch back to position mode, set Bit 4 of PA_1A4 to 0.

4.3 Alarm Reset

Start the alarm reset signal through communication (set Bit 1 of PA_1A4 to 1).

Note: Some drive alarms cannot be reset (for example: encoder line fault, overvoltage fault, undervoltage fault, etc.).

4.4 Absolute positioning control instructions

Method 1: Change the position of the motor rotation by changing the internal position parameters of segment 0, as follows:

Parameter number	Explanation
PA_168	Segment 0 internal position command (32-bit signed data)
PA_169	
PA_190	Segment 0 internal position speed (16-bit unsigned data)

- ① Write the position command parameters PA_168/PA169 and PA_190 via communication.
- ② Start the load position signal (set Bit 5 of PA_1A4 to 1). This will start the motor to rotate to the set position (Note: Bit 5 of PA_1A4 needs to be manually reset to 0).

Method 2: Use the pre-set internal position parameters PA_168~PA_19F (see page 49 of the manual) and then use the position index parameter (PA_091) to select the position to be rotated to, as follows:

Parameter number	Explanation	Setting value
PA_168	Segment 0 internal position command (32-bit signed data)	10000
PA_169		
PA_190	Segment 0 internal position speed (16-bit unsigned data)	100
PA_16A	Segment 1 internal position command (32-bit signed data)	20000
PA_16B		
PA_191	Segment 1 internal position speed (16-bit unsigned data)	100

When PA_091 is set to 0, and then the loading position signal is started (Bit5 of PA_1A4 is set to 1), the motor rotates to the position of 10000 (Note: Bit5 of PA_1A4 needs to be manually changed back to 0);

When PA_091 is set to 1, and then the loading position signal is started (Bit5 of PA_1A4 is set to 1), the motor rotates to the position of 20000 (Note: Bit5 of PA_1A4 needs to be manually changed back to 0).

4.5. Relative positioning control instructions

- ① To use relative positioning, set PA_094 to 0, save the parameters, and restart the drive.
- ② To change the motor's rotational position, modify the internal position parameters in segment 0, as follows:

Parameter number	Explanation
PA_168	Segment 0 internal position command (32-bit signed data)
PA_169	
PA_190	Segment 0 internal position speed (16-bit unsigned data)

- ③. Start the loading position signal (set Bit5 of PA_1A4 to 1), and the motor runs according to the set length. (Note: Bit5 of PA_1A4 needs to be manually changed back to 0);

4.6. Save parameters after power off

Setting the PA_1A7 value to 2049 via communication will save the parameters to the EEPROM. Frequent parameter saving is prohibited, as it may damage the drive.